



Cyber War Heroes

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This is the last line of security defense for the typical Small and Medium Enterprise: The IT Manager. He is often a lone warrior fighting the good fight. She does more than security — she owns the information systems infrastructure across the entire operation. And they are our heroes, leading the cyber-war fight in protecting the information assets of today's small and medium enterprise.

They are asked to deliver new systems, applications, and networks to positively impact their company's bottom line — while being the protectors of the company's most valued possession, it's information assets. They have no choice but to minimise the time and money they spend on security management so they can invest in applications that impact the business — that is until they get hacked, hurt by

some malware, or have sensitive data stolen. Depending on the industry, the large enterprise will often have as many as a dozen or more dedicated security specialists, meanwhile, 87 per cent of SMEs have no one on staff wholly dedicated to security, and spend less than an hour per week proactively managing security.

And so our heroes are those that have figured out the balance between protecting their company's information assets from a cyber-world full of threats, and minimising the time and resource investment so they can focus their efforts on their core business mission.

Increasing Attacks

Our heroes include the likes of Steve Jordan, an IT analyst at Integrated Systems Improvement Service (ISIS) who has a lot more on his plate than only security — he has a network to run and critical applications to support, and so he picks security solutions that allow him to

“spend less time on managing PC security and more time on other tasks.”

Peter Herschel, Director of IT at Gemino Healthcare Finance is a lone cyber-hero at a small healthcare services provider. His organisation is hit each day with the challenges of malicious viruses, phishing scams, spyware, and other threats and so says “I need to keep security risk management simple.” And like our typical cyber-warrior, he'll admit “I have too many other things to do to be an IT security administrator.”

But this is a risky balance to maintain. Each night our last line of defence goes home knowing that they could have spent more time securing their information infrastructure — and thankful that the “big one” didn't hit. They know they are lucky that they weren't attacked because they know the risk. The small and medium enterprise is exposed to the same threats as their large, resource rich enterprise counterparts, and yet their defenses are a fraction of those of the enterprise.

These threats come in several “vectors”, each dangerous in its own way and include the end-point (systems, laptops, and servers), e-mail, Web, and data.

At the end-point (systems, laptops and servers), our own world renowned McAfee AVERT (Anti-Virus Emergency Response Team) Labs report that they've seen more new strains of malware in the past two years than they saw in all of the previous ten — and it's only getting worse. We've identified more new exposures in Q1 of this year than in all of 2007.

Anyone who uses e-mail knows how spam can affect productivity. The popular literature generally estimates between 90-95 per cent of the world's e-mail is spam — for every legitimate e-mail sent, there are between 10-20 pieces of spam. But it is worse than just a productivity drain. Some 25-30 per cent of that spam carries with it some sort of threat — a bot, malware, a link to a malicious site. The implications are huge: If a company receives 3000 legitimate emails today, that means it also received between 30-60K pieces of spam and between 8K and 18K of those contained real and malicious threats.

Data theft and leakage represent the newest and potentially most damaging threat to the small and medium enterprise. 81 per cent of companies surveyed reported the loss of one or more laptops containing sensitive information within the past 12 months. The news is filled with notorious data losses — exposed credit card information, personal ID and social security numbers, and other sensitive data. These reported events are just the tip of the iceberg, and don't include many unreported casualties of lost of company customer records, confidential new designs, or other company secrets.



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Malicious web sites represent the fastest growing threat to a mid-sized business. According to a recent article, nine out of ten websites have some level of exposure to phishing attacks, the cyber equivalent of 'company identity theft'. According to a Gartner Group study from August 2007, nearly 37 per cent of the mid-size companies surveyed had a corporate ID spoofed for phishing purposes. Separately, Forrester Research found that 21 per cent of mid-sized businesses suffered an attack last year, and one in five of these took an entire week to become operational again. And their own websites are often not safe.

New Challenges

While not a threat vector, the burden of regulatory compliance represents a new challenge for the SME, and a demand for precious security-related time and resource. A Forrester study released in February of 2008 indicated of SMEs that had to conform to HIPAA regulations, only 51 per cent really believed they were fully compliant. Only a third in the EU were compliant with the EU Data Protection Directive, and only 32 per cent fully compliant with the Payment Card Industry or PCI Security Standard.

So what do our Cyber Heroes do to confront this challenge? They turn to solutions that are smart, simple and secure. They invest in smart security solutions that have intelligence built in, so that they don't need that same expertise from staff. Since they use these solutions only rarely during the week, they need simple, intuitive functionality that is easy to install, manage and maintain. Done right, the security system should create less work for the lone IT manager, while creating a stronger security environment.

Our Cyber Heroes will never have the security resources of a large enterprise, so they must be smarter in the way they approach security. They must rely on security partners whose products deliver smart, simple and secure information system environments, so they can focus on delivering applications that make a difference. Few of our Cyber Heroes can illustrate this point as well as Scott Gould with the Gynecologic Oncology Group, a cancer research centre, that has a small IT team that needs to stay focused on its core mission, and so chooses solutions that are built to be easy to use and manage so that “we're able to spend much more time doing what we're here to do: stay focused on clinical trials trying to find a cure for cancer.”

We think that's the right priority, too. ■

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